IT6390E - Semantic Web

Exercise 12.9. RDFS - Indicate whether the following statements are true or unknown

Notation: G is the RDF graph given at the beginning of the RDFS exercise. A statement is *true* if it is RDFS-entailed by G, and *unknown* if it is neither entailed nor contradicted.

1.

```
:hasFather rdfs:domain :Person .
```

Unknown. G declares rdfs:domain for :hasParent, not for :hasFather. RDFS entailment contains no rule that passes rdfs:domain from a property down to its sub-properties: rule rdfs2 only uses a domain triple that is already present. Every instance of :hasFather does end up being a :Person, but the domain triple itself is not derivable, so the statement is not entailed.

2.

```
rdfs:range rdf:type rdfs:Resource .
```

True. rdfs:range is the subject of RDFS axiomatic triples such as rdfs:range rdfs:domain rdf:Property, and rule rdfs4a states that the subject of any triple is an instance of rdfs:Resource. More generally, in every RDFS interpretation the extension of rdfs:Resource is the whole universe, so the statement holds independently of G.

3.

```
:hasFather rdfs:range :Father .
```

Unknown. No range is declared for :hasFather in G, and ranges are not inherited from a super-property to a sub-property. Even if every value of :hasFather happens to be a father, RDFS cannot derive the range triple.

4.

```
:hasFather rdfs:domain [ rdfs:subClassOf :Person ] .
```

Unknown. The blank node asks whether there is some class C with :hasFather rdfs:domain C and C rdfs:subClassOf :Person. Since no rdfs:domain triple for :hasFather can be derived at all (see statement 1), there is no candidate for C, so the pattern cannot be instantiated.

5.

```
:Father rdfs:subClassOf [ rdfs:subClassOf :Person ] .
```

True. Take :Person itself as the value of the blank node. :Father rdfs:subClassOf :Person is given in G, and rule rdfs10 makes every class a sub-class of itself, hence :Person rdfs:subClassOf :Person. Both triples of the pattern are therefore entailed and the blank node can be matched.

Summary: 1 unknown, 2 true, 3 unknown, 4 unknown, 5 true.